

1.5 Coordinate Geometry

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs c1–c3

What this topic covers. Straight lines and their gradients, the equation of a circle together with the circle properties MEI expects you to use, and parametric equations — including converting a parametric description into Cartesian form.

1. Straight lines

Quantity	Formula
Gradient through (x_1, y_1) and (x_2, y_2)	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Midpoint	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
Distance	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Point-gradient form	$y - y_1 = m(x - x_1)$

Worked example. Find the gradient, midpoint and distance for $A(1, 2)$ and $B(3, 8)$.

$$m = \frac{8 - 2}{3 - 1} = 3, \quad \text{midpoint } (2, 5), \quad |AB| = \sqrt{2^2 + 6^2} = \sqrt{40} = 2\sqrt{10}$$

Parallel and perpendicular

- **Parallel** lines have **equal** gradients.

- **Perpendicular** lines have gradients whose product is -1 : if one has gradient m , the other has $-\frac{1}{m}$.

Worked example. Find the perpendicular bisector of the segment joining $(1, 2)$ and $(5, 6)$.

Midpoint $(3, 4)$. Gradient of the segment $= \frac{6-2}{5-1} = 1$, so the perpendicular gradient is -1 .

$$y - 4 = -1(x - 3) \implies y = -x + 7$$

Vertical and horizontal lines are the exceptions: $y = 5$ has gradient 0, and $x = 3$ has **undefined** gradient. The product rule for perpendicularity does not apply to this pair — they are perpendicular but you cannot multiply their gradients.

Collinearity

Three points are **collinear** if the gradient between successive pairs is the same.

2. The circle

The circle with centre (a, b) and radius r has equation

$$(x - a)^2 + (y - b)^2 = r^2$$

Note the signs: centre $(2, -3)$ gives $(x - 2)^2 + (y + 3)^2 = r^2$.

The expanded form

Circles often appear expanded, as $x^2 + y^2 + 2gx + 2fy + c = 0$. Recover the centre and radius by **completing the square** in both variables.

Worked example. Find the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.

$$(x^2 - 6x) + (y^2 + 4y) = 12$$

$$(x - 3)^2 - 9 + (y + 2)^2 - 4 = 12$$

$$(x - 3)^2 + (y + 2)^2 = 25$$

Centre $(3, -2)$, radius 5.

Worked example. Find the centre and radius of $x^2 + y^2 + 2x - 6y + 1 = 0$.

$$(x + 1)^2 - 1 + (y - 3)^2 - 9 + 1 = 0 \implies (x + 1)^2 + (y - 3)^2 = 9$$

Centre $(-1, 3)$, radius 3.

The right-hand side is r^2 , not r . Reading a radius of 25 off $(x - 3)^2 + (y + 2)^2 = 25$ instead of 5 is one of the most frequent slips in this topic.

Circle from a diameter

Worked example. A circle has diameter from $(1, 2)$ to $(7, 10)$. Find its equation.

The centre is the midpoint, $(4, 6)$. The radius is half the diameter: $|AB| = \sqrt{6^2 + 8^2} = 10$, so $r = 5$.

$$(x - 4)^2 + (y - 6)^2 = 25$$

Circle properties

Property	Use
The angle in a semicircle is 90°	Recognising a right angle when a diameter is involved
The perpendicular from the centre to a chord bisects it	Finding chord lengths and centres
A tangent is perpendicular to the radius at the point of contact	Finding tangent equations

Worked example. Find the tangent to $x^2 + y^2 = 25$ at $(3, 4)$.

The radius to $(3, 4)$ has gradient $\frac{4}{3}$, so the tangent has gradient $-\frac{3}{4}$:

$$y - 4 = -\frac{3}{4}(x - 3) \implies 3x + 4y = 25$$

Line and circle: how many intersections?

Substitute the line into the circle and look at the discriminant of the resulting quadratic — two roots means two intersections, one means tangency, none means the line misses.

Worked example. For what c is $y = x + c$ tangent to $x^2 + y^2 = 8$?

The perpendicular distance from the centre $(0, 0)$ to the line $x - y + c = 0$ must equal the radius $2\sqrt{2}$:

$$\frac{|c|}{\sqrt{1^2 + (-1)^2}} = 2\sqrt{2} \implies \frac{|c|}{\sqrt{2}} = 2\sqrt{2} \implies |c| = 4$$

So $c = \pm 4$.

(The perpendicular-distance formula used here is **not part of the H640 content** — see section 4.

The method the specification expects is the discriminant: substituting $y = x + c$ into $x^2 + y^2 = 8$ gives $2x^2 + 2cx + c^2 - 8 = 0$, and setting $b^2 - 4ac = 4c^2 - 8(c^2 - 8) = 0$ gives $c^2 = 16$, the same answer.)

Worked example. A point $(5, 5)$ and the circle $x^2 + y^2 = 25$: inside, on, or outside?

Compare $5^2 + 5^2 = 50$ with $r^2 = 25$. Since $50 > 25$, the point lies **outside**.

3. Parametric equations

In a **parametric** description, x and y are each given as functions of a third variable, the **parameter** (often t or θ). Each value of the parameter gives one point on the curve.

Parametric form handles curves that are awkward or impossible to write as $y = f(x)$ — for instance a full circle, which fails the one-output-per-input test.

Converting to Cartesian form

Eliminate the parameter. For polynomial parameters, make t the subject of the simpler equation and substitute. For trigonometric parameters, use $\sin^2 \theta + \cos^2 \theta = 1$.

Worked example. $x = 2t, y = t^2 - 1$.

From the first, $t = \frac{x}{2}$. Substituting:

$$y = \left(\frac{x}{2}\right)^2 - 1 = \frac{x^2}{4} - 1$$

a parabola.

Worked example. $x = \cos \theta, y = \sin \theta$.

$$x^2 + y^2 = \cos^2 \theta + \sin^2 \theta = 1$$

the unit circle.

Worked example. $x = 2 + 3 \cos \theta, y = -1 + 3 \sin \theta$.

Rearrange to isolate the trigonometric functions:

$$\cos \theta = \frac{x-2}{3}, \quad \sin \theta = \frac{y+1}{3}$$

Then apply the identity:

$$\left(\frac{x-2}{3}\right)^2 + \left(\frac{y+1}{3}\right)^2 = 1 \implies (x-2)^2 + (y+1)^2 = 9$$

a circle, centre $(2, -1)$, radius 3.

Worked example. $x = 3 \cos t$, $y = 2 \sin t$.

$$\frac{x^2}{9} + \frac{y^2}{4} = \cos^2 t + \sin^2 t = 1$$

an ellipse.

Worked example. $x = t^2$, $y = 2t$.

From the second, $t = \frac{y}{2}$, so $x = \frac{y^2}{4}$, i.e. $y^2 = 4x$ — a parabola opening to the right.

Choose the equation that is **easier to rearrange** for the parameter — usually the linear one. Making t the subject of $y = t^2 - 1$ forces a $\pm$ square root and needless case-splitting.

Gradient from parametric equations

Using the chain rule,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Worked example. For $x = 2t$, $y = t^2$, find $\frac{dy}{dx}$.

$$\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = 2t \quad \implies \quad \frac{dy}{dx} = \frac{2t}{2} = t$$

4. Perpendicular distance from a point to a line

Not examined on H640. This formula is not part of the OCR B (MEI) content, so no question will require it and no mark scheme will credit it on its own. It is kept here because it is a quick shortcut for tangency questions — but the method H640 expects is the *discriminant*: substitute the line into the circle and set $b^2 - 4ac = 0$. Always be able to do it that way.

The distance from (x_0, y_0) to the line $ax + by + c = 0$ is

$$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

Worked example. Find the distance from $(1, 1)$ to $3x + 4y - 10 = 0$.

$$d = \frac{|3 + 4 - 10|}{\sqrt{9 + 16}} = \frac{3}{5}$$

Rearrange the line into the form $ax + by + c = 0$ before using this — applying it to $y = mx + c$ directly gives nonsense.

5. Common pitfalls

- Reading the radius as r^2 instead of r from the circle equation.
- Getting the sign of the centre wrong: $(x + 1)^2$ means the centre has $x = -1$.
- Forgetting the -9 and -4 correction terms when completing the square in both variables.
- Using the gradient product rule on a vertical line, whose gradient is undefined.
- Eliminating the parameter from the harder equation and introducing a spurious $\pm$.
- Writing $\frac{dy}{dx} = \frac{dx/dt}{dy/dt}$ — the fraction is the other way up.
- Using the perpendicular-distance formula without first rearranging the line to $ax + by + c = 0$.